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228512(28)

Dip. in Engg. (Fifth Semester) Examination, Nov.-Dec. 2015

(ET & T Branch)

ANTENNA & MICROWAVE COMMUNICATION

Time Allowed : Three hours

Maximum Marks : 100

Minimum Pass Marks : 35

Note : Attempt all questions, unless mentioned otherwise. Internal choice is given in questions.

1. Describe Ground Wave Propagation, with neat diagram. What is angle of tilt? 3+4+3=10

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2. State and discuss skip distance with neat sketch. Prove that the skip distance D for a given frequency is given by :

$$D = 2h\sqrt{\left(\frac{f_{muf}}{f_c}\right)^2 - 1}$$

where h is the effective height of the reflecting layer and f_c is the critical frequency of the layer. 5+5=10

3. What is the critical frequency for reflection at vertical incidence if the maximum value of electron density is $1{\cdot}24 \times 10^6$ cm^{-3}? 5

Or

Calculate the value of frequency at which an electromagnetic wave must be propagated through the D-region with an index of refraction of 0·5 and an electron density of $3{\cdot}24 \times 10^4$ electrons/m^3. 5

4. Explain reflex Klystron construction with neat sketch. Also explain the bunching process with applegate diagram. 5+5=10

Or

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Explain Parametric amplifier with Manley-Rowe relation, also explain amplification mechanism of a parametric amplifier. 5+5=10

5. List and explain the limitation of conventional tubes. 10

6. Explain the operation of E-H plane tee junction. Why it is reffered as magnetic tee. Derive the scattering matrix for magic tee. 4+2+4=10

7. Explain the method of measurement of VSWR measurement. CSVTUonline.com 5+5=10

Or

A stip line transmission line has a distance of 0·3175 cm between the ground planes. If the diameter of the equivalent circular conductor is 0·0539 cm, determine the characteristics impedance and velocity of propagation if the dielectric constant is 2·32 for the strip line material. 5+5=10

8. Explain the Bolometer techniques for measurement of power. 10

Or

Explain power ratio method for measurement of attenuation 10

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9. What is transmission line? Give basic principle of a transmission lines. Also draw equivalent circuit of it. 2+4+4=10

10. Explain Yagi-Uda antenna with : 2+3+3+2=10

(i) Introduction

(ii) Diagrams with elements

(iii) Radiation pattern

(iv) General characteristics.

Or

Explain Rhombic antenna with : 2+3+3+2=10

(i) Introduction

(ii) Radiation pattern

(iii) Diagram

(iv) Advantage and disadvantage

11. Write short notes on : (any two) 2½+2½=5

(i) Beam width

(ii) Half wave dipole

(iii) Band width

(iv) Radiation pattern

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